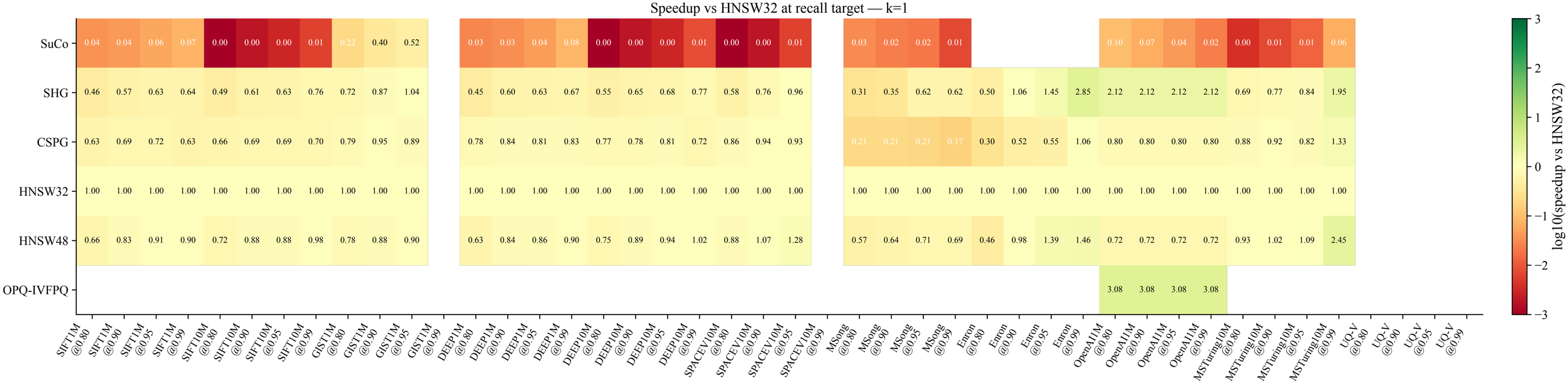
Speedup vs HNSW32 at recall target — k=1
SuCo
SHG
CSPG
HNSW32
HNSW48
OPQ-IVFPQ
0.04 0.04 0.06 0.07 0.00 0.00 0.00 0.01 0.22 0.40 0.52
0.03 0.03 0.04 0.08 0.00 0.00 0.00 0.01 0.00 0.00 0.01
0.03 0.02 0.02 0.01
0.10 0.07 0.04 0.02 0.00 0.01 0.01 0.06
0.46 0.57 0.63 0.64 0.49 0.61 0.63 0.76 0.72 0.87 1.04
0.45 0.60 0.63 0.67 0.55 0.65 0.68 0.77 0.58 0.76 0.96
0.31 0.35 0.62 0.62 0.50 1.06 1.45 2.85 2.12 2.12 2.12 2.12 0.69 0.77 0.84 1.95
0.63 0.69 0.72 0.63 0.66 0.69 0.69 0.70 0.79 0.95 0.89
0.78 0.84 0.81 0.83 0.77 0.78 0.81 0.72 0.86 0.94 0.93
0.21 0.21 0.21 0.17 0.30 0.52 0.55 1.06 0.80 0.80 0.80 0.80 0.88 0.92 0.82 1.33
1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
0.66 0.83 0.91 0.90 0.72 0.88 0.88 0.98 0.78 0.88 0.90
0.63 0.84 0.86 0.90 0.75 0.89 0.94 1.02 0.88 1.07 1.28
0.57 0.64 0.71 0.69 0.46 0.98 1.39 1.46 0.72 0.72 0.72 0.72 0.93 1.02 1.09 2.45
3.08 3.08 3.08 3.08
SIFT1M @0.80
SIFT1M @0.90
SIFT1M @0.95
SIFT1M @0.99
SIFT10M @0.80
SIFT10M @0.90
SIFT10M @0.95
SIFT10M @0.99
GIST1M @0.80
GIST1M @0.90
GIST1M @0.95
GIST1M @0.99
DEEP1M @0.80
DEEP1M @0.90
DEEP1M @0.95
DEEP1M @0.99
DEEP10M @0.80
DEEP10M @0.90
DEEP10M @0.95
DEEP10M @0.99
SPACEV10M @0.80
SPACEV10M @0.90
SPACEV10M @0.95
SPACEV10M @0.99
MSong @0.80
MSong @0.90
MSong @0.95
MSong @0.99
Enron @0.80
Enron @0.90
Enron @0.95
Enron @0.99
OpenAI1M @0.80
OpenAI1M @0.90
OpenAI1M @0.95
OpenAI1M @0.99
MSTuring10M @0.80
MSTuring10M @0.90
MSTuring10M @0.95
MSTuring10M @0.99
UQ-V @0.80
UQ-V @0.90
UQ-V @0.95
UQ-V @0.99
log10(speedup vs HNSW32)
3
2
1
0
−1
−2
−3